

Introduction

We can use the notions of injectivity, surjectivity, and bijectivity to compare the sizes of sets. This is particularly useful in the case of infinite sets! Note that for this conversation, we will revert to using $\mathbb{N} = \mathbb{Z}^*$, as this is the context in which this is usually taught.

Learning Objectives

- Define the inverse of a function and compute inverses for simple functions.
- Describe the relationships between bijective functions and invertibility. Prove that relationship.
- Define the cardinality of a set and what it means for $|S| \leq |T|$ for sets S and T .
- Prove straightforward cardinality relations, such as $|S| \leq |T|$ or $|S| = |T|$.
- Explain which of our number sets $(\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R})$ are of the same size, and why.
- State the Cantor-Schröder-Bernstein Theorem and explain what it means at a high level.
- State and prove the (generalized) Cantor Theorem.

The Inverse of a Function

The word “inverse” has many different meanings depending on the context in which it is used. For example, what if we were to ask you to find the inverse of the number 2? What does this mean? To what are we taking the inverse? To properly understand this, we need to understand the following: Given a binary operator (an operator which takes in two things and produces a single thing in return, such as addition and multiplication of real numbers), we say that a number id is the *identity* of that operator if operating against it does nothing to the input. In the case of addition of real numbers, the operator will satisfy $x + \text{id}_+ = x$ for all possible x ; for example,

$$2 + \text{id}_+ = 2, \quad -5 + \text{id}_+ = -5.$$

Our experience tells us that $\text{id}_+ = 0$. Similarly, for multiplication the identity $\text{id}_\times$ will satisfy $x \times \text{id}_\times = x$ for all x ; for example,

$$3 \times \text{id}_\times = 3, \quad \pi \times \text{id}_\times = \pi.$$

Again our experience tells us that $\text{id}_\times = 1$. We say that 0 is the *additive identity* and 1 is the *multiplicative identity*.

Given an operator and an identity, we say that the *inverse of x* is an element which, when paired against x , gives the identity. The additive inverse of 2 is the

number y such that $2 + y = \text{id}_+ = 0$. In this case $y = -2$, and more generally the additive inverse of n is $-n$. For multiplication, we can convince ourselves that the multiplicative inverse of x is $1/x$; for example, $2 \times (1/2) = 1 = \text{id}_\times$.

Notice that every real number has an additive inverse, while there is no multiplicative inverse for the number 0. In general, one cannot be guaranteed that an inverse always exists for a generic binary operator.

Definition 1

If A, B, C are sets, $f : B \rightarrow C$, and $g : A \rightarrow B$ are functions, we define the *composition* of f and g to be the function

$$(f \circ g) : A \rightarrow C, \quad x \mapsto f(g(x)).$$

If $f, g : A \rightarrow A$, then function composition $f \circ g$ is another example of a binary operator. What is the identity for this operation? Well, we would like a function $\text{id}_\circ : A \rightarrow A$ such that

$$\begin{aligned} f(\text{id}_\circ(x)) &= f(x) \\ &= \text{id}_\circ(f(x)). \end{aligned}$$

The identity function is therefore the function $\text{id}_\circ(x) = x$, the function which does nothing to the argument! Therefore the inverse of a function $f : A \rightarrow A$ is another function $f^{-1} : A \rightarrow A$ such that $f \circ f^{-1} = f^{-1} \circ f = \text{id}_\circ$.

This conversation can be generalized for functions whose domain and codomain are not equal. For example, if $f : A \rightarrow B$ then $f^{-1} : B \rightarrow A$. However, we now require two identities functions, $\text{id}_\circ^A : A \rightarrow A$ and $\text{id}_\circ^B : B \rightarrow B$ such that

$$f^{-1}(f(x)) = \text{id}_\circ^A(x) = x, \quad f(f^{-1}(y)) = \text{id}_\circ^B(y) = y.$$

Definition 2

Let $f : S \rightarrow T$ be a function. We say that $g : T \rightarrow S$ is the *inverse* of f if $f \circ g = \text{id}_\circ^T$ and $g \circ f = \text{id}_\circ^S$. In this case, we say that f is *invertible*.

Example 3

Determine whether each function below is invertible. If so, find an inverse. If not, prove that the function is not invertible.

1. The function $f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x, y) = (2x, x - y)$.
2. The function $g : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$, where $g(n) = \underbrace{100 \cdots 001}_{n\text{-times}} = 10^{n+1} + 1$.
3. The function $h : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ where $h(n)$ returns the smallest prime number p such that $n \leq p$.

Solution.

1. We claim that the function f is indeed invertible, with the inverse given by

$$f^{-1}(z, w) = \left(\frac{z}{2}, \frac{z - 2w}{2} \right).$$

Indeed, by composing these we get

$$f^{-1}(f(x, y)) = f^{-1}(2x, x - y) = \left(\frac{2x}{2}, \frac{2x - 2(x - y)}{2} \right) = (x, y).$$

The other composition is left as an exercise.

2. We claim that this map is not invertible. For the sake of contradiction, suppose a map g^{-1} exists, in which case $g^{-1}(11)$ must be defined and $11 = g(g^{-1}(11)) = 10^{g^{-1}(11)+1} + 1$. This is only possible if $g^{-1}(11) = 0$, which is not a positive integer, giving a contradiction.
3. We claim that this map is not invertible. For the sake of contradiction, suppose a map h^{-1} exists. Note that $h(4) = h(5) = 5$, in which case

$$4 = h^{-1}(h(4)) = h^{-1}(h(5)) = 5,$$

which is obviously not true. Thus h^{-1} cannot exist, and so h is not invertible. ■

Exercise 4 Come up with sets A and B and a function $f : A \rightarrow B$ such that

1. there is a function $g : B \rightarrow A$ such that $f \circ g = \text{id}_B$ but $g \circ f \neq \text{id}_A$.
2. there is a function $g : B \rightarrow A$ such that $g \circ f = \text{id}_A$ but $f \circ g \neq \text{id}_B$.

Relation to Bijectivity

Look back on Example 3, and in particular consider the parts where the function failed to be injective. Take a moment to see if you can determine why these functions failed to be invertible, and if you can connect those failures with something learned in the previous week's readings.

What you hopefully noticed is that Example 3.2 failed to be invertible because it failed to be surjective, while Example 3.3 failed to be invertible because it failed to be injective. It's not too difficult to generalize the arguments given in the solution to Example 3, thereby convincing ourselves that bijectivity is a necessary condition for invertibility. It turns out, bijectivity is also a sufficient condition.

Proposition 5

If $f : S \rightarrow T$ is a function, then f is invertible if and only if f is bijective.

Proof. We will given the ($\Leftarrow$) proof here, and leave the ($\Rightarrow$) direction as an exercise.

Suppose that f bijective and define $g : T \rightarrow S$ as follows: Fix an element $t \in T$ and consider $f^{-1}(\{t\})$, which we know is non-empty as f is surjective. Moreover, $f^{-1}(\{t\})$ consists of a single element, since f is injective. Denote this element by s_t , and define $g(t) = s_t$. We claim that g is the inverse of f .

Indeed, $f(g(t)) = f(s_t) = t$, where in the last equality we've used the fact that $s_t \in f^{-1}(\{t\})$. Similarly, $g(f(s)) = s_{f(s)}$, but since $f(s_{f(s)}) = f(s)$ and f is injective, we get that $s_{f(s)} = s$, as required. $\square$

Exercise 6 Finish the proof of Proposition 5 by showing that an invertible function is necessarily bijective.

Cardinality

Hopefully you've noticed that injectivity and surjectivity can be used to relate the sizes of the domain and codomain. In this sense, they give us an avenue to compare the sizes of sets.

The cardinality of a set S is how many elements are within the set, and is denoted $|S|$. When S is finite, $|S|$ is simple to define; however, the issue becomes trickier when S is an infinite set. For example, we will see that $|\mathbb{N}| = |\mathbb{Z}| = |\mathbb{Q}|$, which is surprising given that it *looks* as though $\mathbb{Q}$ is much larger than $\mathbb{N}$.

Definition 7

Let S and T be sets. We say $|S| \leq |T|$ if there is an injective function $S \rightarrow T$.

For example, if $S = \{1, 2, 3\}$ and $T = \{-3, -6, -9, -12\}$, we could define an injection $f : S \rightarrow T$ by $f(s) = -3s$, showing that $|S| \leq |T|$. This agrees with our usual notion of counting, since $|S| = 3$ and $|T| = 4$. However, we can extend this idea to infinite sets. For example, if $S = \{1, 2, 3, \dots\}$ and $T = \{-1, -2, -3, \dots\}$ then $|S| \leq |T|$ via the injection $s \mapsto -s$. Of course, in this latter case we expect $|S| = |T|$, but we are not yet able to discuss these things.

Before going any further, let's ground our definition in reality.

Proposition 8

If $S = \{s_1, \dots, s_n\}$ and $T = \{t_1, \dots, t_m\}$ are finite sets, then $|S| \leq |T|$ if and only if $n \leq m$.

Proof. $[\Leftarrow]$ Suppose $n \leq m$ and define a map $f : S \rightarrow T$ by $f(s_i) = t_i$. This map makes sense only if $n \leq m$, and moreover it is certainly injective. Hence $|S| \leq |T|$.

$[\Rightarrow]$ By contrapositive, suppose $n > m$. Suppose for the sake of contradiction that an injective function $f : S \rightarrow T$ exists. The data of f includes n outputs, so by the Pigeonhole Principle at least one of these outputs must be repeated; that

is, there exist s_1 and s_2 such that $s_1 \neq s_2$ and $f(s_1) = f(s_2)$, but this contradicts the fact that f is an injection. $\square$

Exercise 9 Show that if $S \subseteq T$ then $|S| \leq |T|$.

The hierarchy of counting thus immediately implies that $|\mathbb{N}| \leq |\mathbb{Z}| \leq |\mathbb{Q}| \leq |\mathbb{R}|$. Similarly, $|[0, 1]| \leq |\mathbb{R}|$. One can also use the notion of a surjection to compare cardinalities, as the following proposition demonstrates:

Proposition 10

If S and T are non-empty sets and $f : S \rightarrow T$ is a surjection, then $|T| \leq |S|$; that is, there is an injection $g : T \rightarrow S$.

Proof. Let $f : S \rightarrow T$ be surjective so that f has a right inverse; namely, there is a function $g : T \rightarrow S$ such that $f \circ g = \text{id}_T$. The symmetry in this relationship means that f is a left-inverse for g , which means that g is injective. Hence we have an injective function from T to S , and $|T| \leq |S|$. $\square$

Once again, we immediately get that if $S = \{s_1, \dots, s_n\}$ and $T = \{t_1, \dots, t_m\}$ are finite sets, then there is a surjection from S to T if and only if $m \leq n$. This leads us to the following definition:

Definition 11

If S and T are sets, then $|S| = |T|$ if there is a bijection $S \rightarrow T$.

If $f : S \rightarrow T$ is a bijection, it is both surjective and injective, so $|S| \leq |T|$ and $|T| \leq |S|$. Thus it makes sense to define equality of cardinality this way. We will also define the strict inequality $|S| < |T|$ if there is an injection from S to T , but no surjection exists.

Note as well that this relationship is immediately transitive, so that if $|A| = |B|$ and $|B| = |C|$, then $|A| = |C|$. Indeed, there is a bijection $f : A \rightarrow B$ and a bijection $g : B \rightarrow C$, then $g \circ f : A \rightarrow C$ is a bijection.

Example 12

Consider the set

$$2\mathbb{N} = \{0, 2, 4, 6, \dots\}.$$

Show that $|2\mathbb{N}| = |\mathbb{N}|$.

Solution. Certainly $2\mathbb{N} \subseteq \mathbb{N}$ so that $|2\mathbb{N}| \leq |\mathbb{N}|$, but we have been asked to go one step further. Define the function $f : \mathbb{N} \rightarrow 2\mathbb{N}$ by $f(n) = 2n$, which we shall show is a bijection.

To see that it is injective, notice that if $f(n) = f(m)$ then $2n = 2m$. Dividing by 2 gives $n = m$ as required. To see that f is surjective, notice that every

positive even number k can be written as $k = 2m$ for some $m \in \mathbb{N}$. Hence $f(m) = 2m = k$ so f is surjective. Since there is a bijection between $\mathbb{N}$ and $2\mathbb{N}$, we conclude that $|\mathbb{N}| = |2\mathbb{N}|$. ■

Example 12 demonstrates that, unlike finite sets, infinite sets can have the same cardinality as their subsets. This is just the first surprising statement in a plethora of unintuitive but interesting results. A similar example is the following, which we've left as an exercise:

Exercise 13 Show that $|(0, 1)| = |\mathbb{R}|$.

Exercise 14 Modify Exercise 13 to show that $|(a, b)| = |\mathbb{R}|$ for any $a < b$. What about the closed interval $[a, b]$?

A subtle question at this point is whether knowing $|S| \leq |T|$ and $|T| \leq |S|$ tells us that $|T| = |S|$. Remember, these “inequalities” are just notation – notation that happens to coincide with our usual intuition for finite sets, but notation nonetheless.

Let's think about this more. We're asking if knowing that there is an injection $f : S \rightarrow T$ ($|S| \leq |T|$) and an injection $g : T \rightarrow S$ ($|T| \leq |S|$) guarantees the existence of a bijection from S to T . That's not at all obvious.

Theorem 15: Cantor-Schröder-Bernstein

If $f : A \rightarrow B$ and $g : B \rightarrow A$ are injective functions, then there is a bijection $h : A \rightarrow B$, and hence $|A| = |B|$.

The proof is somewhat involved, and so is omitted. However, it's worth pointing out that this is not immediately obvious and is difficult to prove.

Example 16

Show that $|(0, 1)| = |[0, 1]|$.

Solution. The inclusion map $\iota : (0, 1) \rightarrow [0, 1], x \mapsto x$ is an injection, so we need only construct an injection in the other direction. Define $f : [0, 1] \rightarrow (0, 1)$ by $f(t) = (1 + 2t)/4$, and note that $f([0, 1]) = [1/4, 3/4]$. This function is injective, for if $f(t_1) = f(t_2)$ then

$$\frac{1 + 2t_1}{4} = \frac{1 + 2t_2}{4} \Rightarrow 1 + 2t_1 = 1 + 2t_2 \Rightarrow t_1 = t_2.$$

Thus by the Cantor-Schröder-Bernstein Theorem, there is a bijection between $(0, 1)$ and $[0, 1]$; hence, $|(0, 1)| = |[0, 1]|$. ■

Definition 17

A set S is said to be *countable* if $|S| \leq |\mathbb{N}|$. Equivalently, S is countable if there is an injective function $f : S \hookrightarrow \mathbb{N}$. We say that S is *countably infinite* if $|S| = |\mathbb{N}|$.

Our goal is to show that $\mathbb{Z}$ and $\mathbb{Q}$ are countable, but that $\mathbb{R}$ is not. For the former, we need the following result:

Proposition 18

A countable union of (pairwise disjoint) countable sets is countable.

Proof. Let $\{A_i\}_{i \in I}$ be a countable collection of pairwise disjoint countable sets, so I is countable, as is each A_i . Let $g : I \hookrightarrow \mathbb{N}$ be an injective function, and for each A_i let $f_i : A_i \hookrightarrow \mathbb{N}$ be an injective function. Define the map

$$f : \bigcup_{i \in I} A_i \rightarrow \mathbb{N}, \quad a \mapsto 2^{g(i)} 3^{f_i(a)}, \quad a \in A_i.$$

This map is well-defined by the uniqueness of prime decompositions and the fact that the A_n are pairwise disjoint. The same uniqueness condition gives injectivity; that is, the only way $2^n 3^m = 2^r 3^s$ is if $n = r$ and $m = s$. Thus a countable union of countable sets is countable. $\square$

Exercise 19 Show that $|\mathbb{N} \times \mathbb{N}| = |\mathbb{N}|$.

Theorem 20

The integers have the same cardinality as the natural numbers: $|\mathbb{Z}| = |\mathbb{N}|$.

Proof. Note that

$$\mathbb{N} \times \mathbb{N} = \{(n, m) : n, m \in \mathbb{N}\} = \bigcup_{n \in \mathbb{N}} \{n\} \times \mathbb{N}$$

is a countable union of countable sets, and hence is countable itself by Proposition 18. Define the map $f : \mathbb{Z} \rightarrow \mathbb{N} \times \mathbb{N}$ as $n \mapsto (|n|, \text{sgn}(n) + 1)$. As an example of what this map does, we have

$$\begin{array}{ll} -5 \mapsto (5, 0) & 5 \mapsto (5, 2) \\ -4 \mapsto (4, 0) & 4 \mapsto (4, 2) \\ -3 \mapsto (3, 0) & 3 \mapsto (3, 2) \\ -2 \mapsto (2, 0) & 2 \mapsto (2, 2) \\ -1 \mapsto (1, 0) & 1 \mapsto (1, 2) \\ 0 \mapsto (0, 1) \end{array}$$

so that the second number just keeps track of whether the number is positive, negative, or zero. We add 1 to the output of the second argument just to ensure

that it's a natural number. Now this map is injective, since if $f(n) = f(m)$ then $(|n|, \text{sgn}(n)) = (|m|, \text{sgn}(m))$. Indeed, equality in the first component, $|n| = |m|$, implies that $n = \pm m$. Equality in the second component, $\text{sgn}(n) = \text{sgn}(m)$, implies that $n = m$.

Since f is injective, we thus have that $|\mathbb{Z}| \leq |\mathbb{N} \times \mathbb{N}| \leq |\mathbb{N}|$. On the other hand, since $\mathbb{N} \subseteq \mathbb{Z}$ it must be that $|\mathbb{N}| \leq |\mathbb{Z}|$. Both inclusions give us that $|\mathbb{N}| = |\mathbb{Z}|$. $\square$

Theorem 21

The rational numbers are countably infinite: $|\mathbb{Q}| = |\mathbb{N}|$.

Proof. Consider the map $f : \mathbb{Q} \rightarrow \mathbb{Z} \times \mathbb{Z}^+$ given by $f(p/q) = (p, q)$ where the fraction p/q is in lowest terms (if the fraction is negative, we always take the sign in the p -component). Again this map is injective, since if $f(p/q) = f(r/s)$ then $(p, q) = (r, s)$ which is true only if $p = r$ and $q = s$. Since $\mathbb{Z}$ and $\mathbb{Z}^+$ are both countable, so too is $\mathbb{Z} \times \mathbb{Z}^+$ and so $|\mathbb{Q}| \leq |\mathbb{Z} \times \mathbb{Z}^+| \leq |\mathbb{N}|$. On the other hand, $\mathbb{N} \subseteq \mathbb{Q}$ so $|\mathbb{N}| \leq |\mathbb{Q}|$ and this gives us that $|\mathbb{Q}| = |\mathbb{N}|$. $\square$

One might expect that this pattern continues forever, and that every infinite set has the same cardinality as the naturals. The real numbers are our first counterexample.

Theorem 22

The real numbers are *strictly larger* than the natural numbers, and so not countable: $|\mathbb{R}| > |\mathbb{N}|$.

Proof. It is sufficient to show that the real numbers $[0, 1]$ are not countable, since then certainly all of $\mathbb{R}$ will be uncountable also. For the sake of contradiction, assume that the real numbers are countable and list them $\{r_1, r_2, r_3, \dots\}$. Write each r_i in its decimal expansion as $r_i = 0.d_1^i d_2^i d_3^i d_4^i \dots$ so that

$$\begin{aligned} r_1 &= 0.d_1^1 d_2^1 d_3^1 d_4^1 \dots \\ r_2 &= 0.d_1^2 d_2^2 d_3^2 d_4^2 \dots \\ r_3 &= 0.d_1^3 d_2^3 d_3^3 d_4^3 \dots \\ r_4 &= 0.d_1^4 d_2^4 d_3^4 d_4^4 \dots \\ &\vdots \end{aligned}$$

Define a new number s as follows. Let $s = 0.s_1 s_2 s_3 s_4 \dots$ where

$$s_i = \begin{cases} 2 & \text{if } d_i^i = 1 \\ 1 & \text{if } d_i^i \neq 1 \end{cases}.$$

The number s is not in the list $\{r_1, r_2, r_3, \dots\}$ by construction (think about this and you will see it is true), but is a real number between 0 and 1. This is a contradiction, since we assumed that we listed all of the real numbers between 0 and 1. We conclude that the real numbers are not countable as required. $\square$

This proof effectively says that given a real number r , it's impossible to decide what the “next” real number should be. Let's say you start at the real number 0. What is the next real number? 0.1? Why not 0.01 or 0.001 or 0.0001? We can put an arbitrary number of zeroes before putting that 1, so it does not make sense to say “the next real number.”

So what goes wrong? Why can't you apply this sort of argument to, say, the rationals? It's tricky to describe without additional work, but the effective answer is that every natural number can be encoded using a finite amount of data. Real numbers on the other hand, generally require an infinite amount of data to explicitly describe. The diagonalization argument above exploits this difference by exhausting the data of the natural numbers without having used the bandwidth of the reals.

Is there a cardinal strictly between $|\mathbb{N}|$ and $|\mathbb{R}|$? This turns out to be an incredibly deep and subtle question, and one that cannot be proven or refuted with our standard set of axioms. One must either *assume* that there is such a cardinal, or *assume* there is no such cardinal, it cannot be proven either way. However, there is a systematic way of taking a set, and getting a set with a strictly larger cardinality.

Definition 23

Let S be a set, and define the power set $\mathcal{P}(S)$ to be the set of all subsets of S . The power set is sometimes denoted 2^S .

Theorem 24: Cantor's Theorem

If S is a set, then $|S| < |\mathcal{P}(S)|$.

Proof. First we show that $|S| \leq |\mathcal{P}(S)|$ by demonstrating an injection $S \rightarrow \mathcal{P}(S)$. Define the function $f : S \rightarrow \mathcal{P}(S)$ by $x \mapsto \{x\}$. This map is evidently injective.

The remainder of the proof is a generalization of the proof given in Theorem 22, and proceeds by diagonalization. Assume that a surjective $f : S \rightarrow \mathcal{P}(S)$ exists and define the set

$$D = \{x \in S : x \notin f(x)\} \in \mathcal{P}(S).$$

We claim D is not the image of any point in S . Indeed, for each element $a \in S$, either $a \in f(a)$ or $a \notin f(a)$. In the first case, if $a \in f(a)$ then $a \notin D$, so $f(a) \neq D$. On the other hand, if $a \notin f(a)$ then $a \in D$, again showing that $D \neq f(a)$. Thus D is not the image of any point, contradicting the assumption that f was a surjection. We conclude that $|S| < |\mathcal{P}(S)|$. $\square$

There is no greatest cardinal, since we can keep taking power sets to create a hierarchy of cardinals. One might be tempted to speak of the set of all sets, but Cantor's theorem shows that such an object cannot exist as a set: if V were the set of all sets, then $\mathcal{P}(V) \subseteq V$, yet Cantor's theorem gives $|V| < |\mathcal{P}(V)|$, a contradiction. The collection of all sets is therefore not a set but a proper class.

Proper classes lie outside the universe of sets and do not carry cardinalities in the usual sense, so they escape the comparison given by Cantor's theorem.

Exercises

1. Determine whether each function below is invertible:

$$(a) \ r : \mathbb{R} \rightarrow \{0, 1\}, r(q) = \begin{cases} 1 & q \in \mathbb{Q} \\ 0 & q \notin \mathbb{Q} \end{cases}.$$

$$(b) \ c : \mathbb{R} \rightarrow \mathbb{R}, c(x) = x^3. \text{ Note: If you want to use calculus for this one, you have to prove whatever theorems you use.}$$

$$(c) \ m : \mathbb{Q} \setminus \{-1\} \rightarrow \mathbb{Q} \setminus \{1\}, m(x) = \frac{x-1}{x+1}.$$

$$(d) \ g : \mathbb{Z} \rightarrow \mathbb{Z}, g(n) = \begin{cases} n/2 & n \text{ even} \\ (n-1)/2 & n \text{ odd} \end{cases}.$$

$$(e) \ s : \mathbb{Q} \rightarrow \mathbb{Z}^+, s(a/b) = |a| + |b|, \text{ where } a/b \text{ is in lowest terms and } b > 0.$$

2. We can generalize Proposition 5 as follows:

(a) A function $f : A \rightarrow B$ is said to be *left-invertible* if there is a function $g : B \rightarrow A$ such that $g \circ f = \text{id}_A$. Show that f is left-invertible if and only if it is injective.

(b) A function $f : A \rightarrow B$ is said to be *right-invertible* if there is a function $g : B \rightarrow A$ such that $f \circ g = \text{id}_B$. Show that f is right-invertible if and only if it is surjective.

3. Show that the following sets have the same cardinality by *explicitly* constructing a bijection. *Note:* You may invoke the existence of known bijections. For example, since we know that $|(0, 1)| = |\mathbb{R}|$, if you want to use a bijection between these sets, you can do so without writing one down.

(a) The negative integers $\mathbb{Z}^-$ and the positive integers $\mathbb{Z}^+$.

(b) $\mathbb{Q}$ and $\mathbb{Q} \cup \{\pi, \sqrt{2}\}$.

(c) The dyadic rationals $D = \{\frac{1}{2^n} : n \in \mathbb{Z}^+\}$ and the rationals $\mathbb{Q}$.

(d) The half open set $[0, 1)$ and the open set $(0, 1)$.

4. Prove, by any means available to you, that the following have the same cardinality.

(a) The irrationals $\mathbb{R} \setminus \mathbb{Q}$ and the reals $\mathbb{R}$.

(b) Let $F = \{A \subseteq \mathbb{Z}^+ : A \text{ is finite}\}$. Show that F is countable.

(c) The set of all polynomials with rational coefficients, denoted $\mathbb{Q}[x]$, is countable.

(d) Show that $\{f : \mathbb{Z}^+ \rightarrow \{0, 1\}\}$ has the same cardinality as $\mathbb{R}$.

5. A real number r is *algebraic* if there exists a polynomial p with rational coefficients such that $p(r) = 0$. Show that the set of algebraic numbers A is countable. *Hint:* You may use the fact that any polynomial can only have finitely many roots, without proof. Exercise 4c will also be useful.